



Optimal plan under Type-II Censoring Schemes in Multi-component Stress-Strength Reliability based on its Estimate and Cost

Somayeh Ghafouri¹, Elham Basiri ²

¹Department of Mathematics, Faculty of Science, Arak University, Arak, 384817758, Iran.

²Department of Mathematics and Applications, Kosar University of Bojnord, Bojnord, Iran.

Abstract:

In this paper, we consider a cut point M^* for number of failure times in the stress component of a multi-component stress-strength system based on progressively Type-II censoring assumed as a truncated geometry distribution while the number of failure times in strength component is assumed fixed. The optimal schemes and the cut points can be considered and the most important is determination them so that the total cost of the test and the estimation of multi-component stress-strength reliability have been reached to minimum and maximum, respectively. A simulation study is carried out to check which different censoring schemes and cut points have the best performance in terms of the total cost and the estimation of multi-component stress-strength reliability. At the end, a real example is given to illustrate the proposed method.

Keywords: Multi-component stress-strength reliability, Type-II right censoring, Truncated geometry distribution.

Mathematics Subject Classification (2010): 62N01, 62N05.

²Corresponding Author: somayyeh.ghafoori@gmail.com

1 Introduction and Preliminaries

In medical, engineering and industrial applications, due to time and cost restrictions in lifetime data, censoring arises when exact lifetimes are only partially known. There are various types of censoring such as Type-I and II right censoring, doubly Type-II censoring, random censoring and progressive Type-I and Type-II censoring, etc. On the other hand, in reliability literature, the stress-strength model described as the reliability of a unit or a system in terms of the stress random variable X and strength random variable Y . The system fails if the stress is greater than the strength, and the probability $R = P(X < Y)$ is the reliability of the system (Kotz et al. (2003)).

A system that consists of more than one component is called a multi-component system. The multi-component stress-strength reliability is defined as the probability that at least s out of k independent and identically distributed (*i.i.d*) strength components survive a common random stress. This model was first introduced by Bhattacharyya and Johnson (1974) and has since been extensively studied due to its wide applicability in fields such as civil engineering, aerospace, and manufacturing Rao (2013).

A combination of Type-II right censoring and multi-component stress-strength reliability can be described as follows. Let k random variables $Z_1, Z_2, \dots, Z_k$ be *i.i.d* random variables with continuous cdf $F_{Z_j}(z_j)$ which are subjected to a common random stress T , an independent random variable with continuous cdf $F_T(t)$.

In recent years, several authors have investigated stress-strength reliability under progressive censoring schemes. For instance, Asgharzadeh et al. (2017) studied estimation of $R = P(X < Y)$ under Type-II right censoring for the Weibull distribution. Jha et al. (2020) considered the problem of estimating multi-component stress-strength (MSS) reliability under progressive Type II censoring when stress and strength variables follow unit Gompertz distributions with common scale parameter. They estimated MSS reliability under frequentist and Bayesian approaches. El-Desokey et al. (2026) developed a reliability framework for multi-component stress-strength systems by modeling stress and strength using the Burr-Hatke distribution under progressive Type II censoring. Maximum likelihood estimators were derived and the stress-strength reliability function, along with their asymptotic confidence intervals.

The novelty of this work lies in assuming that the number of failure times in the strength component is random variable from the truncated geometric distribution.

2 Main Results

Let $Z_{i1:n_1} \leq \dots \leq Z_{ik:n_1}$, $i = 1, \dots, m$, denote a Type-II right censored sample from the exponential distribution with parameter λ_1 , i.e. $f(z_{ij}) = \lambda_1 \exp(-\lambda_1 z_{ij})$; $\lambda_1 > 0$ $j = 1, \dots, k$, under the Type-II right censoring as well as let $T_{1:n_2} \leq \dots \leq T_{m:n_2}$ be another such sample taken from the exponential distribution with parameter λ_2 under the Type-II right censoring. Then the reliability of a multi-component stress-strength model is written as

$$G_{s,k} = P(\text{at least } s \text{ of the } (Z_1, \dots, Z_k) \text{ exceed } T) = \sum_{p=s}^k \binom{k}{p} \int_{-\infty}^{+\infty} [1 - F_Z(t)]^p [F_Z(t)]^{k-p} f_T(t) dt.$$

Here, let k be fixed and M (the number of failures in stress model) be distributed as the truncated geometric distribution at points M^* and 0

$$P(M = m) = \frac{\theta(1 - \theta)^m}{P(1 \leq M \leq M^*)} = \frac{\theta(1 - \theta)^m}{(1 - \theta - (1 - \theta)^{M^*+1})}; \quad m = 1, 2, \dots, M^* \quad (2.1)$$

Then, the likelihood function of λ_1 , λ_2 and θ expressed as

$$L(\lambda_1, \lambda_2, \theta) = f(Z_{i1}, \dots, Z_{ik}) = \sum_{m=1}^{M^*} f(z_{ij}|M = m)P(M = m), \quad (2.2)$$

in which, we know that

$$\begin{aligned} f(z_{ij}|M = m) &= C_1 \prod_{i=1}^m \left[C_2 \prod_{j=1}^k f(Z_{ij})(1 - F(Z_{ik}))^{(n_1-k)} \right] \\ &\times f(t_i) [1 - F(t_m)]^{(n_2-m)}, \quad C_1 = \frac{n_2!}{(n_2 - m)!}, \quad C_2 = \frac{n_1!}{(n_1 - k)!}. \end{aligned} \quad (2.3)$$

In this paper, we first estimate the multi-component stress-strength reliability function by estimating the three parameters λ_1 , λ_2 and θ . Then, we examine the designs that maximize the estimated multi-component reliability function, which is a function of $\hat{\lambda}_1$, $\hat{\lambda}_2$ and $\hat{\theta}$ as well as minimizing the introduced cost function.

Then, the likelihood function of λ_1 , λ_2 and θ is given by

$$\begin{aligned} L(\lambda_1, \lambda_2, \theta) &= \frac{\theta}{(1 - \theta - (1 - \theta)^{M^*+1})} \sum_{m=1}^{M^*} C_1 (1 - \theta)^m C_2^m \lambda_1^{km} \\ &\times e^{(-\lambda_1 [\sum_{i=1}^m \sum_{j=1}^k z_{ij} + (n_1 - k) \sum_{i=1}^m z_{ik}])} \lambda_2^m e^{(-\lambda_2 [\sum_{i=1}^m t_i + m t_m (n_2 - m)])}. \end{aligned} \quad (2.4)$$

After simplification, calculating the first partial derivatives of (2.4) with respect to λ_1 , λ_2 and θ along with equating to zero, the likelihood equations do not have explicit solutions. Therefore, we use the numerical solution to solve them. After getting maximum likelihood estimators $\widehat{\lambda}_1$, $\widehat{\lambda}_2$ and

$\widehat{\theta}$ from likelihood equations, we substitute them in

$$\begin{aligned}
 G_{s,k} &= \text{The probability of stress - strength} = P(\text{at least } s \text{ of the } Z_j > T; j = 1, 2, \dots, k) \\
 &= \sum_{p=s}^k \binom{k}{p} \int_0^{+\infty} [1 - F_Z(t)]^p [F_Z(t)]^{k-p} f_T(t) dt \\
 &= \sum_{p=s}^k \binom{k}{p} \int_0^{+\infty} e^{-\lambda_1 t p} [1 - e^{-\lambda_1 t}]^{k-p} \lambda_2 e^{-\lambda_2 t} dt \\
 &= \sum_{p=s}^k \binom{k}{p} \int_0^{+\infty} \lambda_2 e^{-(\lambda_2 + \lambda_1 p)t} [1 - e^{-\lambda_1 t}]^{k-p} dt,
 \end{aligned} \tag{2.5}$$

and we can get $\widehat{G}_{s,k}$ which is a function of M^* , $\widehat{\lambda}_1$, $\widehat{\lambda}_2$ and $\widehat{\theta}$. Then, the $\widehat{G}_{s,k}$ can be calculated.

The total cost function can be rewritten as (see Basiri and Beigi (2020))

$$E(T_C) = C_0 + C_t E(T_{m:n_2}) + C_u n_2, \tag{2.6}$$

which C_0 is the cost of setting up a test and other related costs as well as C_t and C_u are time-dependent cost on test and cost per unit under test, respectively. Also, $T_{m:n_2}$ is m^{th} (last) order statistic of $(T_{1:n_2}, \dots, T_{m:n_2})$ from Type-II right censored sample taken under the exponential distribution with parameter λ_2 parameter.

In order to compute $E(T_{m:n_2})$, let $f(t_i) = \lambda_2 e^{-\lambda_2 t_i}$; $\lambda_2 > 0$, by appropriate transformations (see David and Nagaraja (2003)), we can easily see that $T_{i:n_2} = \sum_{j=1}^i \frac{z_j}{\lambda_2(n_2-j+1)}$ and $Z_i; i = 1, 2, \dots, m$ are all independent and identically distributed as standard exponential.

Therefore,

$$E(T_{i:n_2}) = \sum_{j=1}^i \frac{1}{\lambda_2(n_2-j+1)}.$$

As a result,

$$E(T_C|M=m) = C_0 + C_t \sum_{j=1}^m \frac{1}{\lambda_2(n_2-j+1)} + C_u E(n_2|M=m). \tag{2.7}$$

Finally, by double expectation theorem,

$$\begin{aligned}
 \text{Cost} = E_M[E(T_C|M=m)] &= C_0 + \frac{C_t}{\lambda_2} \sum_{m=1}^{M^*} \sum_{j=1}^m \frac{\theta(1-\theta)^m}{(n_2-j+1)(1-\theta-(1-\theta)^{M^*+1})} + \\
 &\quad \left[\sum_{m=1}^{M^*} \frac{m\theta(1-\theta)^m}{(1-\theta-(1-\theta)^{M^*+1})} + \sum_{m=1}^{M^*} \frac{(n_2-m)\theta(1-\theta)^m}{(1-\theta-(1-\theta)^{M^*+1})} \right] C_u.
 \end{aligned} \tag{2.8}$$

The optimal censoring plan in Type-II right censoring and optimal M^* based on minimum of final cost function (2.8) and maximum of $\widehat{G}_{s,k}$ calculated by (2.5) is considered.

3 Simulation Study

In this section, the results of previous section are presented. We consider $C_0 = 20$; $C_t = 3$; $C_u = 12$. Based on 3 different schemes of Type-II right censored sample with size m taken from the exponential distribution with λ_2 and 3 other schemes of Type-II right censored sample with size k taken from the exponential distribution with λ_1 , which censored sample is optimal in terms of minimizing cost function and maximizing of $\widehat{G}_{s,k}$.

Algorithm 1. This algorithm has the following steps:

- Step 1: First, for given primary and arbitrary values of the parameter λ_2 from the exponential distribution a Type-II right censored sample with sample size m is generated.
- Step 2: Again, for given primary and arbitrary values of the parameter λ_1 from the exponential distribution, a Type-II right censored sample with different sample sizes of k is generated.
- Step 3: Using "optim" in R, compute the maximum likelihood estimates of λ_1 , λ_2 and θ .
- Step 4: By substitution of $\widehat{\lambda}_1$, $\widehat{\lambda}_2$ and $\widehat{\theta}$ get $\widehat{G}_{s,k}$.
- Step 5: After calculation of the $\widehat{G}_{s,k}$, acquire the objective function $Obj = \frac{\widehat{G}_{s,k}}{Cost}$ (see page 236 from [Pakdaman and Ahmadi \(2015\)](#)).
- Step 6: Replicate the above processes 1000 times and report the best M^* that has minimum objective function for different quantities of l , s for some Type-II censored samples.

4 Tables and Figures

This section contains Tables 1 to 3. The results obtained from Table 3 are as follows:

- (1) In Table 3 for $\lambda_1 = 0.5$; $\lambda_2 = 7$, as s increases in all schemes, the objective function Obj decreases. Then, in order to increase the objective function, it is better to choose small values for s .

Table 1: Various Type-II right censored samples with size $m = 1, \dots, M^*$.

Case of F	
1F	$n_2 - m = 2$
2F	$n_2 - m = 5$
3F	$n_2 - m = 10$

Table 2: Various Type-II right censored samples with size k , $k = 7, 10, 15$.

Case of G	
1G	$n_1 - k = n_1 - 7 = 16$
2G	$n_1 - k = n_1 - 10 = 7$
3G	$n_1 - k = n_1 - 15 = 19$

(2) From Table 3, in all schemes, under $\lambda_1 = 0.5$; $\lambda_2 = 7$, for $s = k$, it can be seen the largest reduction in the objective function. So, in order to decrease the value of the objective function, it is better to choose $s = k$.

(3) In Table 3, it can be roughly said that under different Type-II right censoring schemes, the M^* remains minimum and fixed for different s 's.

Discussion and Conclusion

In this paper, we introduced the concept of determining an optimal value for M^* , the cut point of the number of failure times in the stress component of a multi-component stress-strength system based on Type-II censoring assumed as truncated geometry distribution while the number of failure times in strength component is assumed fixed. Also, the optimal scheme along with cut points can be considered and are determined in such a way that the total cost of the test and the estimation of multi-component stress-strength reliability have been reached to minimum and maximum, respectively. A simulation study results that in order to control M^* , different s 's for special censoring schemes can be applied. This method can be generalized to some other censoring types such as progressive Type-II censoring schemes etc.

Table 3: the best M^* along with minimum quantity of the objective function for 3 Type-II censored samples with sizes m (stress distribution) and k (strength distribution) for different s , $\lambda_1 = 0.5$; $\lambda_2 = 7$, $C_0 = 20$, $C_t = 3$ and $C_u = 12$ for $B = 1000$ replications.

Case G		1F	2F	3F
G1	$s = 3$	$5.841 \times 10^{-5} (M^* = 3)$	$6.927 \times 10^{-5} (M^* = 3)$	$7.545 \times 10^{-5} (M^* = 3)$
	$s = 4$	$4.164 \times 10^{-5} (M^* = 3)$	$4.939 \times 10^{-5} (M^* = 3)$	$5.379 \times 10^{-5} (M^* = 3)$
	$s = 6$	$1.753 \times 10^{-5} (M^* = 3)$	$2.079 \times 10^{-5} (M^* = 3)$	$2.264 \times 10^{-5} (M^* = 3)$
	$s = 7$	$8.194 \times 10^{-6} (M^* = 3)$	$9.718 \times 10^{-6} (M^* = 3)$	$1.058 \times 10^{-6} (M^* = 3)$
G2	$s = 4$	$5.109 \times 10^{-5} (M^* = 3)$	$6.059 \times 10^{-5} (M^* = 3)$	$6.600 \times 10^{-5} (M^* = 3)$
	$s = 5$	$4.616 \times 10^{-5} (M^* = 3)$	$5.474 \times 10^{-5} (M^* = 3)$	$5.963 \times 10^{-5} (M^* = 3)$
	$s = 9$	$1.208 \times 10^{-5} (M^* = 3)$	$1.432 \times 10^{-5} (M^* = 3)$	$1.560 \times 10^{-5} (M^* = 3)$
	$s = 10$	$5.770 \times 10^{-6} (M^* = 3)$	$6.843 \times 10^{-6} (M^* = 3)$	$7.454 \times 10^{-6} (M^* = 3)$
G3	$s = 3$	$5.390 \times 10^{-5} (M^* = 3)$	$6.393 \times 10^{-5} (M^* = 3)$	$6.963 \times 10^{-5} (M^* = 3)$
	$s = 6$	$5.646 \times 10^{-5} (M^* = 3)$	$5.510 \times 10^{-5} (M^* = 3)$	$6.001 \times 10^{-5} (M^* = 3)$
	$s = 9$	$3.736 \times 10^{-5} (M^* = 3)$	$4.431 \times 10^{-5} (M^* = 3)$	$4.826 \times 10^{-5} (M^* = 3)$
	$s = 12$	$1.691 \times 10^{-5} (M^* = 3)$	$2.000 \times 10^{-5} (M^* = 3)$	$2.184 \times 10^{-5} (M^* = 3)$
	$s = 15$	$3.865 \times 10^{-6} (M^* = 3)$	$4.584 \times 10^{-6} (M^* = 3)$	$4.993 \times 10^{-6} (M^* = 3)$

References

- Asgharzadeh, A., Valiollahi, R. and Raqab, M. Z. (2017). Estimation of $P(X < Y)$ for Weibull Distribution under Progressive Type-II Censoring. Communications in Statistics - Simulation and Computation, 46(6), 4477–4496.
- Basiri, B. and Beigi, S. (2020). The optimal scheme in type II progressive censoring with random removals for the Rayleigh distribution based on two-sample Bayesian prediction and cost function. Journal of advanced Mathematical Modeling, 10(12), 135–157.
- Bhattacharyya, G. K. and Johnson, R. A. (1974). Estimation of Reliability in a Multicomponent Stress-Strength Model. Journal of the American Statistical Association, 69(348), 966–970.
- David, H. N. and Nagaraja, H. N. (2003). Order Statistics. Third Ed., John Wiley & Sons: Hoboken, NJ.
- El-Desokey, R., El-Awady, M. M., Ahmad, H. H. and El-Gohary, A. (2026). Multicomponent Stress-Strength Reliability Inference under Progressive Type-II Censoring: A One-Parameter Model with Applications to Power Systems and Earthquake Data. AIMS Mathematics, 11(3), 8031–8064.
- Jha, M. K., Dey, S., Alotaibi, R. M. and Tripathi, Y. M. (2020). Reliability Estimation of a Multicomponent Stress-Strength Model for Unit Gompertz Distribution under Progressive Type II Censoring. Quality and Reliability Engineering International, 36(3), 965–987.

- Kotz, S., Lumelskii, Y. and Pensky, M. (2003). The Stress-Strength Model and its Generalizations: Theory and Applications. World Scientific: Singapore.
- Pakdaman, Z. and Ahmadi, J. (2019). Switching Time of the Standby Component to the k-out-of-n:G System in the Stress-Strength Setup. *Metrika*, 82(2), 225–248.
- Rao, G. S. (2013). Estimation of Reliability in Multicomponent Stress-Strength Model for Inverse Rayleigh Distribution. *International Journal of Reliability and Applications*, 14(1), 1–12.